

Dhvani Rakshak — Ultimate Master Defense PDF Guide

Complete Technical, Real-World, Regulatory & Presentation Q&A; Collection

■ Part 1: The 30-Second Opening Pitch

"Cybercriminals use AI voice cloning tools (like ElevenLabs & VALL-E) to impersonate bank executives and IAS/IPS officers, stealing crores over phone calls. Standard IVR systems cannot stop this because the caller sounds identical to the victim."

Dhvani Rakshak is India's first real-time, multi-layered AI voice clone defense system. It scans live calls in under **200 milliseconds**, detecting acoustic anomalies and phase disruptions to automatically cut scam calls—all while maintaining **100% compliance with India's DPDP Act 2023 by saving zero raw audio on disk.**

■ Part 2: Technical & Architecture Defense Q&As;

Frontend Architecture & UI Tech Stack

Q1: Why did you choose React.js and Vite over Next.js or create-react-app?

Ans: Vite uses native ES Modules (ESM) and Rollup, giving us sub-second HMR and a lightweight production bundle (<300 KB) essential for low-latency dashboard streaming.

Q2: How is state managed across the 10 different feature modules without performance lag?

Ans: *App.jsx* handles active tab navigation state while each feature component manages localized hooks, preventing unnecessary re-renders across unmounted views.

Q3: How did you implement the dark glassmorphism theme without external frameworks like Tailwind?

Ans: Built a custom CSS design system using HSL color tokens (*var(--text-muted)*), CSS Grid/Flexbox, and hardware-accelerated *backdrop-filter: blur(12px)*.

Q4: How does the UI render live audio waveforms dynamically on the screen?

Ans: Using HTML5 Web Audio API *AudioContext* and SVG dynamic canvas elements to plot frequency spectrum bars in real time.

Audio Signal Processing & Feature Extraction

Q5: What exact audio preprocessing steps are applied before feature extraction?

Ans: Raw audio PCM bytes are resampled and normalized to a **16 kHz mono float32 array** with amplitudes scaled between -1.0 and +1.0.

Q6: How does sliding window chunking work for live call streaming?

Ans: Audio is sliced into **500 ms sliding windows with a 100 ms hop size**, producing continuous risk evaluations every 100 milliseconds.

Q7: What exact 40 acoustic features are extracted from the voice signal?

Ans: **40 Mel-Frequency Cepstral Coefficients (MFCCs)**, pitch jitter percentage, shimmer dB, spectral roll-off, spectral centroid, and zero-crossing rates.

Q8: How do you mathematically detect an ElevenLabs or deepfake voice clone?

Ans: AI voices exhibit **artificially flat pitch jitter (< 0.05%)**, high-frequency phase gaps (> 4 kHz), and zero biological breathing pauses.

Security, Cryptography & Privacy

Q9: How do you enforce a 'Zero Raw Audio Retention' policy under India's DPDP Act 2023?

Ans: Audio frames are processed strictly in volatile RAM memory buffers and immediately flushed (*del raw_data*) as soon as feature extraction completes.

Q10: How are authentic voice prints stored securely during registration?

Ans: Converted into a **192-dimensional numerical vector** (d-vector / ECAPA-TDNN) encrypted with **AES-256**, making original speech reconstruction impossible.

Q11: Why do you use SHA-256 hashing for call audit logs?

Ans: SHA-256 creates a unique 256-bit cryptographic digest for every detection event, guaranteeing tamper-evident audit logs for regulatory compliance.

Q12: How does Polygon Blockchain provide legal non-repudiation in court?

Ans: We write the SHA-256 Merkle root of the voice vector to a Polygon smart contract, creating an immutable public timestamp that cannot be edited by anyone.

Explainable AI, Performance & Scaling

Q13: How does your Explainable AI (XAI) engine calculate feature importance?

Ans: Uses **SHAP (SHapley Additive exPlanations)** to score which specific acoustic metrics (pitch, shimmer, spectrum) contributed most to raising the threat score.

Q14: How does the system handle multi-modal risk calculation?

Ans: Computes a weighted risk score: $\text{Total Risk} = 0.5 * \text{Voice Acoustics} + 0.3 * \text{Behavioral Dynamics} + 0.2 * \text{Context}$.

Q15: How would the backend scale to handle 10,000 concurrent audio WebSockets?

Ans: Using **Docker microservices**, Redis Pub/Sub async audio queues, and ONNX Runtime with TensorRT GPU model acceleration.

Q16: How do you prevent false positives when a user has a cold or sore throat?

Ans: Sore throats alter vocal timbre but **preserve natural human pitch micro-vibrations**. Borderline scores (50-70%) trigger step-up OTP checks instead of a hard call cut.

■ Part 3: Real-World, Banking & Business Defense Q&As;

Problem & Real-World Necessity

Q1: Why does India need Dhvani Rakshak right now when banks already use OTPs?

Ans: OTPs verify *what you have*, not *who is speaking*. Scammers use realistic AI voice clones over live calls to trick victims into sharing OTPs or approving wire transfers voluntarily.

Q2: Who suffers the most from AI voice clone scams in the real world?

Ans: Bank account holders, elderly citizens, corporate finance managers, and IAS/IPS VIPs targeted for deepfake extortion and wire fraud.

Q3: What is the financial scale of this problem in Indian banking?

Ans: Voice scam losses run into hundreds of crores annually, with single executive impersonation frauds exceeding **■50 Lakhs to ■2 Crores**.

Q4: How does Dhvani Rakshak stop fraud before money leaves the bank?

Ans: By analyzing live call audio in **under 200 ms** and automatically cutting the line or triggering OTP step-up verification before the transaction is authorized.

Practical Deployment & Telecom Reality

Q5: How will a bank actually install Dhvani Rakshak into their call centers?

Ans: As a lightweight **SIP/RTP Media Gateway sidecar** that taps into existing PBX switches (Cisco/Avaya) via WebSockets without replacing legacy systems.

Q6: How do you handle poor 2G/4G cellular network coverage or background noise?

Ans: We normalize speech to **16kHz mono** and track fundamental pitch phase consistency (F0), which stays stable even through lossy mobile network compression.

Q7: What happens when 100,000 customers call a bank simultaneously during peak hours?

Ans: Our backend uses **Docker microservices**, Redis Pub/Sub async audio queues, and GPU model acceleration to handle thousands of concurrent calls seamlessly.

Q8: How will you commercialize or deploy this project in the market?

Ans: As a **B2B SaaS Security API** for banks, IVR providers, and government defense desks on a per-call subscription model.

■ Part 4: Presentation Key Metrics Cheat Sheet

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- **Processing Latency:** < 200 ms (Sub-second real-time call interception)
 - **Voice Embedding Dimension:** 192-dimensional numerical vector (ECAPA-TDNN / d-vector)
 - **Acoustic Features Extracted:** 40 MFCCs, Pitch Jitter (%), Shimmer (dB), Spectral Roll-off, Phase Discontinuity
 - **Security & Privacy Encryption:** AES-256 for voice vector embeddings, SHA-256 for session audit hashes
 - **Legal Non-Repudiation:** Polygon Blockchain Smart Contract (SHA-256 Merkle Root)
 - **Compliance Badges:** India DPDP Act 2023, RBI Cyber Security Framework 2023, GDPR Compliant (Zero Raw Audio Saved)